

EPGP Academic Scheduling Optimization using Linear Programming

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Introduction

This formulation represents the scheduling problem for the IIM EPGP Program, accommodating different sections/areas, courses, time slots, and faculty members. The objective is to create an optimal schedule that satisfies all hard constraints while minimizing violations of a configurable consecutive-period soft constraint.

1 Sets and Indices

- $b \in B$: Set of buckets. Buckets are either section identifiers (in Sections mode) or area short-names (in Areas mode).
- $c \in C$: Set of Courses.
- $t \in T_b$: Set of time slots available for bucket b .
- $f \in F$: Set of Faculty members.
- $d \in D$: Set of unique Calendar Dates in the quarter.
- $pk \in PK_b$: Set of **period keys** for bucket b — unique integers that identify each scheduling period. The encoding is determined by the configured period unit:

- `period_unit = "weeks"`: $pk = \text{iso_year} \times 100 + \text{iso_week}$ (e.g. Week 10 of 2026 $\rightarrow$ 202610)
- `period_unit = "days"`: $pk = \text{date.toordinal}()$ (unique integer per calendar day)

Two period keys pk and pk' are **consecutive** if they represent immediately adjacent periods — successive ISO weeks (with correct year-rollover handling) or successive calendar days. Note that *arithmetic* $pk + 1$ is **not** the same as the consecutive successor for week-encoded keys (e.g. 202652 $\rightarrow$ 202701, not 202653). We write $\text{succ}^j(pk_0)$ for the j -th consecutive successor of pk_0 .

- $g \in G$: Set of course conflict groups (used by optional constraint $H7$).
- $C_g \subseteq C$: Set of courses in conflict group g .
- $B_g \subseteq B$: Set of buckets to which conflict group g applies.

2 Parameters

- $R_{b,c}$: Required number of sessions for bucket b in course c .
- $\text{Faculty}(b, c)$: The faculty member assigned to teach course c in bucket b .
- $\text{Date}(b, t)$: The calendar date associated with slot t for bucket b .
- $\text{Period}(b, t)$: The period key pk associated with slot t for bucket b (see PK_b above).

- **Boundary(b, t):** The boundary-group key of slot t for bucket b . Controls where the consecutive counter resets:
 - `reset_boundary = "month"`: $\text{Boundary}(s, t) = \text{year} \times 100 + \text{month}$
 - `reset_boundary = "none"`: $\text{Boundary}(s, t) = 0$ (no reset; sequence is continuous)
- **Unavailable(f, d):** Binary flag; 1 if faculty f is unavailable on date d , 0 otherwise. *(used by optional constraint H6)*
- M_f : Maximum sessions faculty f may teach per calendar day.
- M : Maximum number of **consecutive periods** in which the same course may be taught to the same scope without incurring a penalty (user-configured; default $M = 2$). A window of $M + 1$ consecutive same-boundary periods that all have active sessions constitutes one violation.
- **Scope:** The consecutive-rule and auxiliary variables operate over a configurable *scope*:
 - Sections mode: $\text{scope} = (b, c)$ (bucket-course pair, i.e., per-section behavior)
 - Areas mode: $\text{scope} = c$ (course-level scope aggregated across buckets/areas)

3 Decision Variables

- $x_{b,c,t} \in \{0, 1\}$: 1 if bucket b is assigned course c at time slot t ; 0 otherwise. (In Sections mode this enforces one course per (b, t) ; in Areas mode multiple concurrent $x_{b,c,t} = 1$ for the same (b, t) are allowed unless forbidden by other constraints.)
- $y_{\text{scope}, pk} \in \{0, 1\}$: Auxiliary variable; 1 if the given *scope* has at least one session of the relevant course in period pk ; 0 otherwise.
- $p_{\text{scope}, pk_0} \in \{0, 1\}$: Penalty variable; 1 if the scope is active in *every* period of the window $(pk_0, \text{succ}(pk_0), \dots, \text{succ}^M(pk_0))$; 0 otherwise. Created only when the consecutive rule is enabled.

4 Objective Function

The objective depends on whether the consecutive sessions rule is enabled in the configuration.

Case 1 — Consecutive rule enabled (Soft):

$$\text{Minimize } Z = \sum_{s \in S} \sum_{c \in C} \sum_{pk_0} p_{s,c,pk_0} \quad (1)$$

Minimize the total number of windows in which the same course is taught in $M + 1$ or more consecutive same-boundary periods for the same section.

Case 2 — Consecutive rule disabled (Pure feasibility):

$$\text{Minimize } Z = 0 \quad (2)$$

No soft objective; the solver simply finds any schedule satisfying all active hard constraints.

5 Constraints

Fixed Hard Constraints (always applied)

- **H1: Session Requirement**

$$\sum_{t \in T_b} x_{b,c,t} = R_{b,c} \quad \forall b \in B, c \in C \quad (3)$$

Every bucket must complete the exact required number of sessions for every assigned course.

- **H2: One Course per Slot (Zero Slack)**

$$\sum_{c \in C} x_{b,c,t} = 1 \quad \forall b \in B, t \in T_b \quad (\text{Sections mode only}) \quad (4)$$

In Sections mode each available time slot for a bucket (section) must be occupied by exactly one course. This constraint is not applied in Areas mode, where areas may host parallel sessions in the same slot.

- **H3: No Faculty Cloning**

$$\sum_{\substack{(b,c): \\ \text{Faculty}(b,c)=f \\ \text{Date}(b,t)=d, \text{FromTime}(b,t)=\tau}} x_{b,c,t} \leq 1 \quad \forall f \in F, (d, \tau) \quad (5)$$

A faculty member cannot teach more than one bucket at the same date and start time.

- **H4: Maximum Daily Workload**

$$\sum_{b \in B} \sum_{c \in C} \sum_{\substack{t \in T_b: \\ \text{Date}(b,t)=d, \\ \text{Faculty}(b,c)=f}} x_{b,c,t} \leq M_f \quad \forall f \in F, d \in D \quad (6)$$

A faculty member may teach at most M_f sessions per calendar date across all buckets.

- **H5: Course Spacing (Daily)**

$$\sum_{\substack{t \in T_b: \\ \text{Date}(b,t)=d}} x_{b,c,t} \leq 1 \quad \forall b \in B, c \in C, d \in D \quad (7)$$

A bucket cannot be scheduled for the same course more than once on the same calendar day.

Optional Hard Constraints (can be disabled via configuration)

- **H6: Faculty Unavailability** (*Optional*)

$$x_{b,c,t} = 0 \quad \text{if } \text{Unavailable}(\text{Faculty}(b,c), \text{Date}(b,t)) = 1 \quad \forall b \in B, c \in C, t \in T_b \quad (8)$$

A class cannot be scheduled on a date when the assigned faculty member is marked unavailable.

- **H7: Course Conflict Groups (Bucket Scope)** (*Optional*)

$$\sum_{b \in B_g} \sum_{c \in C_g} \sum_{\substack{t: \\ \text{Date}(b,t)=d, \\ \text{FromTime}(b,t)=\tau}} x_{b,c,t} \leq 1 \quad \forall g \in G, (d, \tau) \quad (9)$$

For each conflict group g , at most one course from C_g may run at any given date and start time across the applicable buckets B_g .

6 Soft Constraint: Consecutive Sessions Rule

When the consecutive rule is enabled, we penalize every window in which the same course is taught to the same section in $M + 1$ or more consecutive same-boundary periods. The formulation uses auxiliary period-activity indicators y to link slot-level decisions x to the period-level penalty variables p .

Window Eligibility

A window $\mathcal{W} = (pk_0, \text{succ}(pk_0), \dots, \text{succ}^M(pk_0))$ of length $M + 1$ over (s, c) is **eligible** if and only if:

1. **Coverage:** All $M + 1$ period keys in $\mathcal{W}$ exist in PK_s for section s with valid slots mapped to course c .
2. **Consecutiveness:** Each successive pair in $\mathcal{W}$ are adjacent periods (next ISO week, or next calendar day), including correct year-boundary handling.
3. **Boundary confinement** (when `reset_boundary` $\neq$ "none"):

$$\text{Boundary}(s, t_0) = \text{Boundary}(s, t_j) \quad \forall j = 1, \dots, M \quad (10)$$

where t_0, t_j are representative slots in pk_0 and $\text{succ}^j(pk_0)$ respectively. This ensures the consecutive counter resets at month (or other configured) boundaries, so violations cannot span across them.

Auxiliary Constraints (linking y to x)

- **A1: Period Activity — Lower Bound**

$$y_{\text{scope}, pk} \geq x_{b,c,t} \quad \forall b \in B, c \in C, t \in T_b \text{ where } \text{Period}(b, t) = pk \text{ and scope applies} \quad (11)$$

$y_{\text{scope}, pk}$ is forced to 1 if *any* session within the scope is scheduled in period pk .

- **A2: Period Activity — Upper Bound**

$$y_{\text{scope}, pk} \leq \sum_{\substack{t \in T_b \\ : \text{Period}(b, t) = pk \\ \text{and scope applies}}} x_{b,c,t} \quad \forall pk \in \bigcup_b PK_b \quad (12)$$

$y_{\text{scope}, pk}$ is forced to 0 if *no* session within the scope is scheduled in period pk .

Penalty Activation

- **P: Consecutive Period Penalty**

For every eligible window $\mathcal{W} = (pk_0, \text{succ}^1(pk_0), \dots, \text{succ}^M(pk_0))$ over the chosen scope:

$$p_{\text{scope}, pk_0} \geq \sum_{pk \in \mathcal{W}} y_{\text{scope}, pk} - M \quad (13)$$

Activation logic:

- If every period in $\mathcal{W}$ is active ($y = 1$ for all $M + 1$ periods): $\sum_{pk \in \mathcal{W}} y_{\text{scope}, pk} = M + 1$, so RHS = 1, forcing $p_{\text{scope}, pk_0} \geq 1$ — the penalty fires.
- If at most M periods are active: $\sum_{pk \in \mathcal{W}} y_{\text{scope}, pk} \leq M$, so RHS ≤ 0 . Since $p \geq 0$ always holds, the constraint is slack and p remains 0.

Special cases recovered by this general formulation:

- $M = 1$, `reset_boundary` = "none": reduces to the classic *Alternate-Week Rule* — the same course cannot be taught in two consecutive weeks within the configured scope.
- $M = 2$, `reset_boundary` = "month": implements the *Two-Consecutive-Week Rule within a Month* — within the chosen scope the course may appear in at most 2 consecutive weeks per month; a third consecutive week in the same month triggers a penalty.

Implementation Notes

The solver implementation generalizes the original per-section model by introducing a bucket abstraction and a configurable *scope* for the consecutive-rule. Key points:

- The decision variables and constraints are constructed over buckets B so the same code supports Sections and Areas modes.
- The consecutive-rule auxiliary variables y and penalties p are indexed by the chosen scope (either (b, c) pairs or course-level c) so penalties are computed correctly in each mode.
- Helper routines compute period adjacency and boundary keys robustly across year boundaries (ISO week rollovers) to ensure correct window eligibility checks.